

Machine Learning Based Framework for Cyberbullying and Harassment Detection in Online Platform

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Abstract—Cyberbullying and online harassment have become serious problems with the rapid growth of social media platforms. People of all ages are exposed to bullying comments, hateful speech, and targeted harassment, which can cause emotional distress and even long-term psychological harm. This paper presents a machine learning based framework for detecting cyberbullying and harassment in online platforms. The proposed system uses text processing and a Support Vector Machine (SVM) classifier to identify whether a given message is bullying or non-bullying. In addition, the framework includes a legal support mechanism that can guide users about possible legal actions when extreme or repeated abuse is detected. Experiments were conducted using real Twitter data sets, and the system achieved an accuracy of about 92%, demonstrating that the model is effective and suitable for real-world deployment.

Index Terms—Cyberbullying Detection, Online Harassment, Machine Learning, Support Vector Machine, Text Classification, Social Media.

I. INTRODUCTION

The rapid expansion of social networking platforms such as Twitter, Instagram, Facebook, and Reddit has revolutionized online communication by enabling real-time interaction, information sharing, and digital collaboration on a global scale. While these platforms offer significant social and educational benefits, they have also created environments where bullying and harmful behavior can spread rapidly. Cyberbullying and online harassment have emerged as major challenges in the digital era, affecting users across different age groups, particularly adolescents and young adults. Kowalski, Limber, and Agatston highlighted that cyberbullying differs from traditional bullying because of anonymity, continuous exposure, and large-scale dissemination through digital platforms. Similarly, Patchin and Hinduja demonstrated the strong relationship between cyberbullying and psychological issues such as anxiety, depression, and low self-esteem among victims [2], [7].

The increasing volume of user-generated content on social media platforms makes manual monitoring and moderation extremely difficult. Conventional rule-based filtering systems are often ineffective because harmful messages may contain slang, sarcasm, coded language, or context-dependent expressions that are difficult to detect using keyword matching alone. Dinakar et al. [7] proposed one of the early intelligent cyberbullying detection systems using contextual reasoning and semantic analysis, emphasizing the limitations of simple keyword-based approaches. Furthermore, Al-Garadi et al. [2] demonstrated the use of machine learning methods for cybercrime and cyberbullying detection on Twitter data. These studies motivated the adoption of Natural Language Processing (NLP) and Machine Learning (ML) techniques for automated cyberbullying detection.

To address these challenges, this work proposes a machine learning-based framework for cyberbullying detection and reporting. The proposed system applies NLP preprocessing techniques such as tokenization, stopword removal, stemming, and TF-IDF vectorization to transform textual data into meaningful numerical representations. Multiple supervised learning algorithms—including Logistic Regression, Support Vector Machine (SVM), and Naïve Bayes—are utilized for classification. Davidson et al. [6] demonstrated the effectiveness of supervised learning algorithms for detecting offensive language, while Cortes and Vapnik [4] established Support Vector Machines as a powerful classification technique for high-dimensional text data. The models are trained and validated on labeled data sets containing social media comments categorized as bullying or non-bullying content, achieving accuracies of up to 0.93 and F1-scores above 0.90. Additionally, the framework incorporates a legal reporting and evidence documentation model to support victims facing repeated online bullying. However, for cases involving serious online bullying,

the existing automatic detection approaches cannot guarantee a high level of accuracy. The gap between automatically detecting severe cyberbullying and providing concrete legal support is filled by our proposed approach.

II. RELATED WORK

The increasing proliferation of online communication and social media networks has generated considerable interest in the field of cyberbullying detection. In the earlier days, keyword-based methods and user-defined lexicons were employed to identify inappropriate or bullying messages. Although these were simple to implement, they suffered from a lack of sophistication in detecting disguised cyberbullying such as sarcasm, slang and implicit requests. Subsequently, researches gradually adopted machine learning techniques for automatic text classification. For instance, Davidson et al. [6] provided a clear indication that machine learning techniques such as Logistic Regression and SVM are efficient in classifying hate speech and offensive messages using linguistic patterns derived from labeled data. In addition, Cortes and Vapnik [4] were pioneers in introducing the Support Vector Machine algorithm for classifying data in high-dimensional spaces. More recently, Chatzakou et al. [3] identified the importance of combining textual and behavioral analyses in detecting aggressive online communication over Twitter.

To cope with the problem, researchers focused on adding feature-based and context-based methods for higher performance. Dadvar et al. [5] confirmed that integrating user-related features significantly increases classification accuracy. Zhao et al. [12] and Xu et al. [10] established that semantic relationships, conversational context and linguistic patterns play an essential role in discriminating bullying messages from normal conversations on social media. Van Hee et al. [9] proposed methods for automatic detection of cyberbullying with the use of linguistic and context-based features in the social media domain and revealed the limitations of a pure lexical approach and the need for feature engineering based on context.

Deep learning models are recently explored to capture hidden semantic representations and relationships of the bullying messages. For instance, Zhang et al. [11] utilized a CNN-GRU hybrid architecture to learning local text patterns and sequential relationships in hate speech detection. Agrawal and Awekar [1] applied deep learning models on various online communities and datasets and demonstrated their efficacy in cyberbullying detection. Literature survey was carried out by Salawu et al. [8] on automatic detection of cyberbullying, which illustrated the progression from traditional machine learning models to sophisticated deep learning systems. The proposed system is different from existing works in that not only are machine learning models used for effectively detecting severe cyberbullying, but legal support and reporting facilities for the victims are also provided.

III. METHODOLOGY

The cyberbullying detection system as presented in the pipeline aims at classifying the online text in an appropriate manner. Tweets are collected from the public data set [url is cited here in reference to the public data set¹ which has cyberbullying and non-cyberbullying tweets with labels. The data set covers many types of abuse and hence the model is not just able to capture different patterns of cyber bullying behavior but also effective in separating the clean tweets from them.

TABLE I
DATA SET DISTRIBUTION

Classes	Number of Tweets
Normal Tweets	1000
Cyberbullying Tweets	1000
Total Tweets	2000

As shown in table I shows the distribution of the data set. This data set has 2000 tweets in total which is evenly distributed in cyberbullying and normal categories. Balanced data set is important to avoid bias during training and have a dependable classification result. Cyberbullying class contains the tweets that are abusive, hateful, bullying and threaten tone while normal class represents the social media interaction that is neutral and not bullying. There are a wide range of writing styles, slang, abbreviations, emotions, emoticons and informal slang words included in this data set which can better represent social media conversations and challenge the classification task which enables the proposed framework perform better.

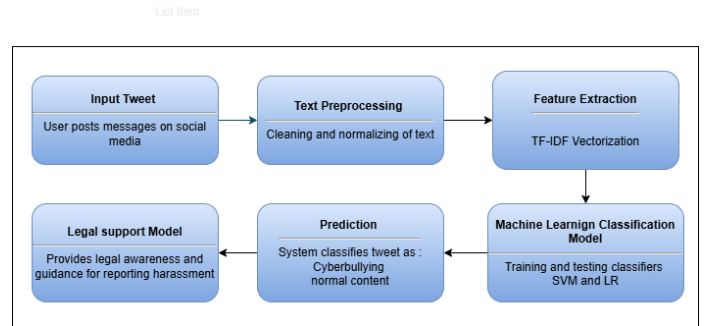


Fig. 1. System Architecture of the Proposed Cyberbullying Detection Framework

A. System Architecture

Figure 1 provides the framework architecture for the cyberbullying detection system that we proposed. The system takes input tweets which are collected from the social media, then goes to the text preprocessing stage. Here we eliminate the characters like special character, URLs, unnecessary text and standardize the input.

¹<https://www.kaggle.com/datasets/andrewmvd/cyberbullying-classification>

After this, we will extract the features from the preprocessed text by using TF-IDF (Term Frequency-Inverse Document Frequency) method to represent the text as numerical features. These features will be given to machine learning models, such as SVM and Logistic Regression for the prediction of classifying tweets into cyberbullying or non-cyberbullying. Accordingly to the output prediction the system will identify bullying tweets, if it continues for sometime or more harmful it will be activating a legal awareness module which will make users aware about relevant laws and procedure of reporting and how to deal with it.

B. Data Collection

Tweets are collected from publicly available datasets containing labeled examples of both cyberbullying and non-cyberbullying content. The dataset includes different categories of online harassment, such as religion, ethnicity, age, and personal attacks, along with normal social media interactions. These labeled instances provide the ground truth required for supervised machine learning and help the model learn patterns associated with bullying and non-bullying behavior.

C. Data Balancing

A balanced distribution of cyberbullying and non-cyberbullying tweets is maintained. So that the algorithm does not have any favor to a particular class, we ensure the balance of dataset. Balancing the dataset makes the classification more fair and accurate by ensuring the model doesn't bias towards the majority class. Also it helps in generalizing and performing consistent predictions on future social media data.

D. Text Preprocessing

After the collection of tweets, these tweets go through several preprocessing and normalization stages in order to produce accurate data and constant while training models. All the tweets are initially converted to lower case so that words with varying casing are treated the same. Noise from the tweets are removed including URLs, #tags, tags, punctuation, emoticons, numbers, special characters, etc. To keep only informative words of the tweet. Additionally, common English stop words like 'the', 'is', 'and', etc. are removed, to reduce insignificant words from the tweet.

Tweets are then tokenized ²(split into words or tokens), then the words are either lemmatized or stemmed, to keep only the root form of the word. The additional white spaces and character repetitions are removed during the preprocessing steps. By performing the preprocessing steps on the collected tweets, the amount of noise from the tweet is reduced and the features are improved resulting in better model generalization of the proposed classification.

²<https://www.geeksforgeeks.org/nlp/what-is-tokenization/>

E. Feature Extraction

Preprocessed textual data is then converted into numerical feature vectors using TF-IDF (Term Frequency-Inverse Document Frequency) technique³. This method assigns scores to words depending on how many times they are mentioned in a tweet and how significant they are overall and the model picks out all the relevant terms required for detecting cyberbullying.

$$TF-IDF(t, d) = TF(t, d) \times \log \left(\frac{N}{DF(t)} \right) \quad (1)$$

This method is formulated by equation (1). The frequency of the words in one document and inversely frequency in corpus will be assigned to word by TF-IDF. It also enhances meaningful words representation in text classification. In equation (1), $TF(t, d)$ is term t document d , which represents frequency of term t in document d . $DF(t)$ is frequency of term t over document corpus which represents the number of documents that term t contain. N represents the number of documents that all over corpus and also defines number of documents in corpus which have this term t in it. Those words which are more informative and less frequently appear would receive higher weights. Thus TF-IDF would effectively select a good word to detect cyberbullying.

F. Dataset Splitting

To test the effectiveness of the suggested cyberbullying detection system, the dataset is split into an 80% training set and a 20% testing set. The training data is used to calculate model parameters, learn patterns of bullying and non-bullying data, and test data is used to determine the accuracy of the model with unknown data. The division is used to allow for accurate evaluation of the model's capacity for generalization, and also to minimize the likelihood of overfitting.

G. Model Training

The classification task is carried out mainly using the Support Vector Machine (SVM) classifier whereas Logistic Regression (LR) is used as a comparative baseline model. SVM performs best with high dimensional text data from TF-IDF feature extraction as it tries to find a hyperplane which gives the best separation between the cyberbullying and the non-cyberbullying classes. Logistic Regression helps in finding the probabilities for classification and also evaluating the traditional ML algorithm in relation to detecting bullying content.

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w \cdot x_i + b) \geq 1 \quad (2)$$

Equation (2) defines the optimization goal of the SVM classifier where it tries to discover an optimal hyperplane such that the margin between classes is maximized. The weight vector is labeled as w , bias term is b , and the class label of the training data is y_i . By optimizing the margin

³<https://www.geeksforgeeks.org/machine-learning/understanding-tf-idf-term-frequency-inverse-document-frequency/>

to maximize classification accuracy the SVM provides high classification performance on unknown textual data, thereby ensuring improved generalization performance.

The classification is made using the sigmoid activation function which squashes the input to values in the range [0, 1] representing the probability of belonging to a class, and classify as cyberbullying/non-cyberbullying accordingly. Logistic Regression is used as a good baseline model as it is simple and effective on binary classification problems.

$$P(y = 1|x) = \frac{1}{1 + e^{-(w \cdot x + b)}} \quad (3)$$

Equation (3) computes the probability of a data sample belonging to the positive class using the sigmoid function, where w and b denote the model parameters.

H. Model Evaluation

The trained models are evaluated using standard performance metrics such as precision, recall, and F1-score.

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

Equation (4) defines precision, which measures the proportion of correctly predicted positive observations among all predicted positive observations.

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

Equation (5) represents recall, which measures the proportion of correctly identified positive observations among all actual positive observations.

$$F1 - score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (6)$$

Equation (6) defines the F1-score as the harmonic mean of precision and recall, providing a balanced measure of classification performance. Here, TP , FP , and FN represent True Positives, False Positives, and False Negatives, respectively.

I. Prediction and Classification

For the predictions the test tweets are processed through the same preprocessing and TF-IDF pipeline, and the trained model predicts whether the content is cyberbullying or non-cyberbullying.

J. Legal Awareness Model

If repeated or severe bullying content is detected, the system activates a legal support module that provides guidance on relevant cyber laws and reporting procedures. This enhances the practical usability of the system beyond detection.

IV. RESULTS AND EVALUATION

To determine the efficiency of the cyberbullying detection model, Twitter data sets including real life cyberbullying and non cyberbullying messages have been used. The data has been divided into training and testing data set in 80:20 ratio, and Machine learning classifiers (SVM and LR) are trained and validated over unknown data.

It is observed that the accuracy of the cyberbullying detection is 92.95% using Support Vector Machine (SVM) classifier and 87.10% using Logistic Regression (LR), which means the model can discriminate between cyberbullying and non-cyberbullying effectively. The confusion matrix reveals that, 949 normal tweets and 910 bullying tweets are predicted and only a small number of false tweets are observed to confirm the robust predictive ability of the model.

More precisely, based on classification report, precision is found to be 0.95, recall to be 0.91 and f1 score for cyberbullying class is calculated as 0.93, so that there is a good balance between correctly classified as bully and errors produced for this class. Further on, ROC curve has also given AUC of 0.99 showing clear separation between the classes, therefore the proposed framework has good accuracy and is robust enough for cyberbullying detection.

A. Confusion Matrix Description

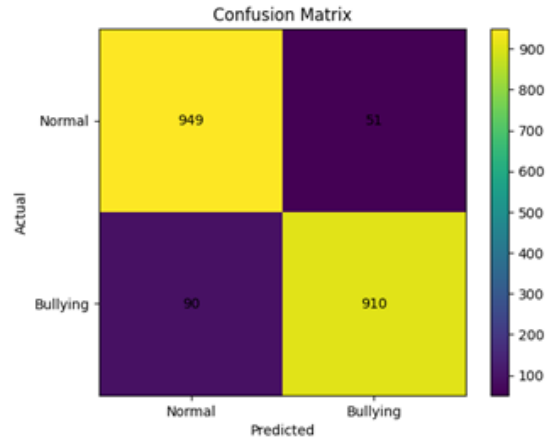


Fig. 2. Confusion Matrix of the Proposed Model (SVM)

From the confusion matrix shown in Fig. 2 we can view the performance result of proposed SVM for detecting cyberbullying. 949 from 1000 normal tweets are classified as non-cyberbullying tweet and 51 normal tweets are wrongly classified as cyberbullying tweet. Besides, among 1000 cyberbullying tweets, 910 of them are correctly classified as harmful tweet while 90 tweets of them are misclassified as normal tweets. It can be noticed that the system could achieve the performance of distinguishing cyberbullying tweet from normal tweet in a rather well performance with low misclassification rate and high prediction reliability in real life application.

B. Model Performance Analysis

TABLE II
PERFORMANCE OF THE PROPOSED MODELS

Model Name	Accuracy (%)	F1-Score	Precision
SVM	92.95%	0.95	0.95
LR	87.10%	0.87	0.89

Table II displays the comparative analysis of performance among the machine learning models applied for cyberbullying detection. Support Vector Machine (SVM) classifier outperformed the other models in terms of accuracy (92.95%), F1 score (0.95) and precision (0.95). It demonstrates consistent and excellent classification performance as Logistic Regression (LR) model performs at 87.10% accuracy with an F1 score of 0.87 and a precision of 0.89. SVM proves to be the more appropriate approach for detecting cyberbullying compared to LR with its ability to handle large dimensionality of TF-IDF features as well as classifying between cyberbullying and non-cyberbullying text based on the margin it creates..

C. ROC Curve

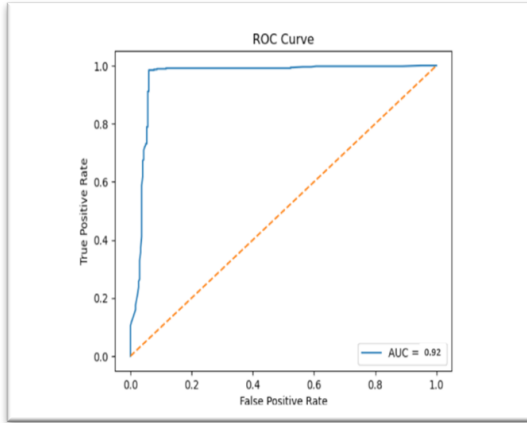


Fig. 3. ROC Curve

Figure 3 provides the ROC curve for our suggested cyberbullying detection model, which describes the relationship between the true positive rate and the false positive rate for the model. The value of Area Under the Curve (AUC) is 0.92, which suggests good performance of classification and shows how well the classification model works to differentiate cyberbullying tweet from non-cyberbullying tweets. A large AUC value implies that the proposed model works robustly and reliably.

Overall Performance Explanation: The proposed framework TF-IDF + SVM performs effectively on the detection of cyberbullying in social media as listed in Table III. The model can achieve an accuracy of 92.95% while also having higher precision, recall, F1-score, and ROC-AUC values, thus implying reliable performance of classification. The excellent

TABLE III
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Metric	TF-IDF + SVM	TF-IDF + Logistic Regression
Accuracy	92.95%	87.00%
Precision	0.93	0.87
Recall	0.93	0.87
F1-Score	0.93	0.87
ROC-AUC	0.94	0.88

value of the recall indicate that it is able to find out almost all bullying related comments and it also able to reduce some false positive detection with higher precision.

Error Analysis: About 90 tweets labeled as bullying were incorrectly classified because they use of sarcasm, low-level of bullying words, indirect harassment, multi-language tweets, or context dependent words. These difficulties show that traditional ML methods have limitation in recognizing underlying semantic sense and conversation context available in social media language.

Model Comparison: As shown in Table III, TF-IDF + SVM has the best accuracy of 92.95%. In comparison, TF-IDF + Logistic Regression got about 87% accuracy. The reason that SVM performed better than Logistic Regression could be that SVM has strong ability of finding margin based separation in high-dimensional and sparse text feature spaces, hence better for cyberbullying detection.

V. CONCLUSION AND FUTURE WORK

The paper described a machine learning based system for detecting cyber bullying and harassment in social networks. The proposed system uses the TF-IDF technique for feature extraction, and Support Vector Machine for prediction. The system shows significant accuracy of about 92%, proving its efficacy for detecting bullying content in real-world data sets. In addition to the classification accuracy of the system, the inclusion of a legal counseling module increases the practical applicability by enabling users to seek assistance in understanding potential legal consequences. This system has potential to create a more secure and responsible internet by automatically flagging bullying content with a legal perspective.

In terms of future work, one can improve the proposed system by using social network API's to achieve real time classification of the reported contents. Transformer based approaches such as Multilingual BERT could be considered to enhance feature extraction to understand the contextual meanings of text. Such systems could possibly identify sarcasm, implicature speech and code switching or mixed language tweets. Additionally, expanding the number of languages could improve the system's performance. These modifications would increase the robustness, scalability, and real-world usefulness of the current system for detecting cyber bullying.

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